

In this assignment, your task is to implement the A^* algorithm and investigate the impact of heuristic quality on search performance. In particular, your task is to reproduce the experimental results of Russell and Norvig in the 8-puzzle domain (see sections 3.6–3.6.1 in the supplemental PDF on Moodle that contains a scan of the relevant pages from their book *Artificial Intelligence: A Modern Approach* (3rd Edition)). In particular, your task is to reproduce the table shown in figure 3.29.

A few things to note:

- In the description of their experiments, Russell and Norvig state that they “generated 1200 random problems”. How are you going to generate random instances of the 8-puzzle problem (i.e., random starting configuration)? Simply picking a random position for each tile will not work — it turns out that the state space for the 8-puzzle actually comprises two disconnected components¹. If your starting position happens to be drawn from the wrong component, then there will be no path that leads you to the goal state and the puzzle will be unsolvable.
- Some program design advice: begin by writing a class to encapsulate the state and behavior of the puzzle board. Implement the A^* algorithm itself in a separate module using a collection of functions/static methods.
- In your final presentation, you should compare the results from your experiments to those carried out by Russell and Norvig. Do you find that they agree? If not, be sure to address why you believe there is a discrepancy.

Your primary deliverable for this assignment is a 15–20 minute presentation with prepared slides. You should submit your slides before the deadline via Moodle. You should also invite me as a collaborator to your GitHub repository containing all the code you wrote for this assignment; I will consider the last commit to the `main` branch before the deadline as the final version of your code for the purposes of assessment.

In addition to the two heuristics discussed in the scanned pages from the book, implement a third, admissible heuristic from the literature on heuristic search and compare its performance to h_1 and h_2 . The following paper suggests one heuristic, and includes references to other papers that discuss other heuristics, and would be a good starting point for your research: <http://academiccommons.columbia.edu/catalog/ac%3A141289>

¹This was first pointed out in a paper about 140 years ago! See the paper by Johnson and Story: <https://www.jstor.org/stable/2369492>

Recommended Timetable

Here's a recommendation for how to budget your time over the next couple of weeks as you work on this assignment.

- **Sep. 5–9:** Read the experiment description from Russell and Norvig, think about how to design the 8-puzzle board class, write and debug your code for representing and manipulating an 8-puzzle board, think about how you would generate random puzzle start states and how to solve for the effective branching factor.
- **Sep. 10–13:** Implement and debug A^* , collect data and compare it to Russell and Norvig's results, think about and analyze any discrepancies, create the *Introduction* and *Background* sections of your slide deck.
- **Sep. 14–18:** Complete your experiments, take a step back and think about your presentation's narrative, create drafts of the *Experiments* and *Results* sections of your presentation. Check-ins with Dr. Ramanujan about the status of your code, experiments and presentation draft during class time.
- **Sep. 19–23:** Wrap-up any pending experiments, finalize your slide deck.